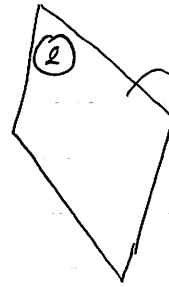
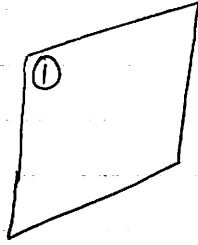


View Factors and Blackbody Exchange



Blackbody Surfaces

$$\left. \begin{aligned} q_{1 \rightarrow 2} &= F_{12} A_1 e_{b1} \\ q_{2 \rightarrow 1} &= F_{21} A_2 e_{b2} \end{aligned} \right\}$$

F_{ab} = view factor = fraction of radiation leaving surface "a" and reaching surface "b".

So our net exchange is:

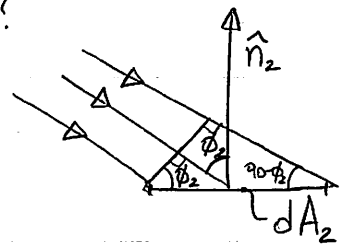
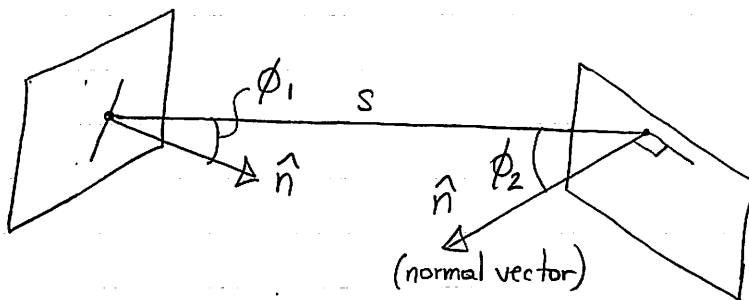
$$q_{12} = q_{1 \rightarrow 2} - q_{2 \rightarrow 1} = F_{12} A_1 e_{b1} - F_{21} A_2 e_{b2}$$

If $T_1 = T_2$, $e_{b1} = e_{b2}$

$$q_{12} = e_b (A_1 F_{12} - A_2 F_{21}) = 0 \quad (\text{no net energy transfer between bodies at same } T)$$

$$\therefore \boxed{A_1 F_{12} = A_2 F_{21}}$$

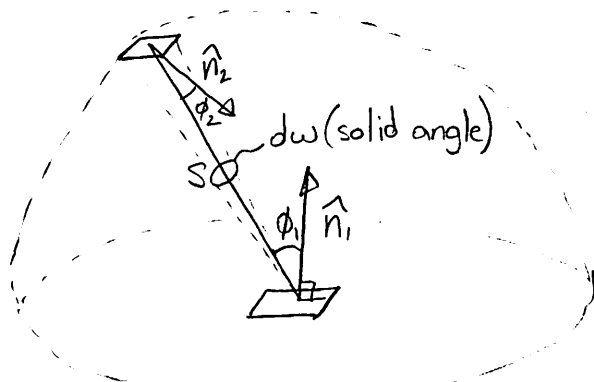
But what is our fraction F_{12} or F_{21} ?



$$A_1 F_{12} = B \int_{A_1} \int_{A_2} \frac{\cos \phi_1 \cos \phi_2}{s^2} dA_1 dA_2 ; B = \text{constant}$$

Aside:

Think of the previous expression in terms of small differential areas:



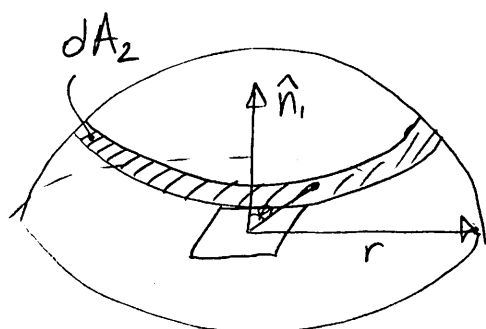
$$d\omega = \frac{\cos \phi_2 dA_2}{s^2} \text{ (Solid Angle)}$$

$$E_{\text{out}} = I_{\text{out}} \cos \phi_1 dA_1 d\omega$$

$$I_{\text{out}} = \frac{\sigma T^4}{\pi} \text{ (Intensity)}$$

You can solve the rest. Try it!

Going a step further



$$A_1 = dA_1$$

$$s = r$$

$$\phi_2 = 0$$

$$dA_2 = 2\pi r^2 \sin \phi_1 d\phi_1$$

Substituting back into our relation: $\underbrace{A_1}_{dA_1} F_{12} = B \int_{A_1} \int_{A_2} \frac{\cos \phi_1 \cos \phi_2}{s^2} dA_1 dA_2$

$$F_{12} = B \cdot 2\pi \int_0^{\pi/2} \cos \phi_1 \sin \phi_1 d\phi_1$$

$$F_{12} = 1$$

$$1 = B \cdot 2\pi \int_0^{\pi/2} \sin \phi_1 d(\sin \phi_1) = B \cdot 2\pi \left[\frac{(\sin \phi_1)^2}{2} \right]_0^{\pi/2} = \pi B = 1$$

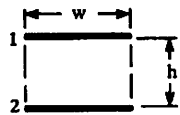
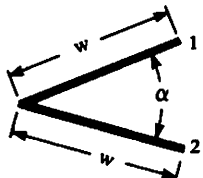
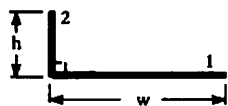
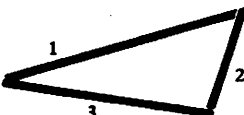
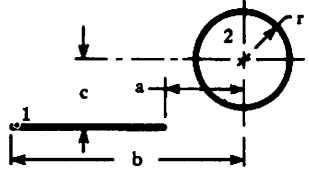
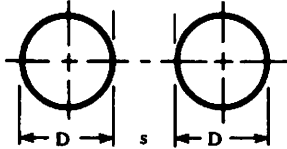
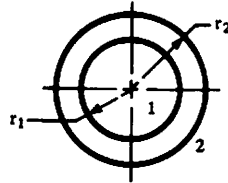
$$B = \frac{1}{\pi}$$

So our view factor becomes:

$$A_1 F_{12} = A_2 F_{21} = \frac{1}{\pi} \int_{A_1} \int_{A_2} \frac{\cos \phi_1 \cos \phi_2}{s^2} dA_1 dA_2$$

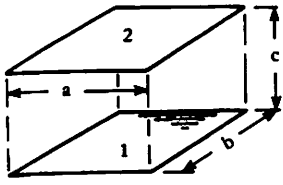
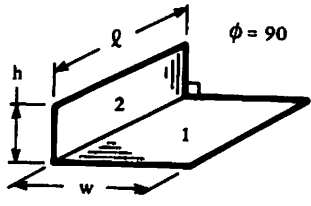
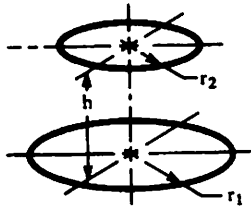
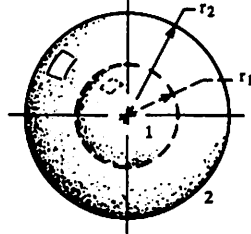
⇒ Look up in Mills Table 6.1, pg. 541-543.

Table 10.2 View factors for a variety of two-dimensional configurations (infinite in extent normal to the paper)

Configuration	Equation
1. 	$F_{1-2} = F_{2-1} = \sqrt{1 + \left(\frac{h}{w}\right)^2} - \left(\frac{h}{w}\right)$
2. 	$F_{1-2} = F_{2-1} = 1 - \sin(\alpha/2)$
3. 	$F_{1-2} = \frac{1}{2} \left[1 + \frac{h}{w} - \sqrt{1 + \left(\frac{h}{w}\right)^2} \right]$
4. 	$F_{1-2} = (A_1 + A_2 - A_3)/2A_1$
5. 	$F_{1-2} = \frac{r}{b-a} \left[\tan^{-1} \frac{b}{c} - \tan^{-1} \frac{a}{c} \right]$
6. 	Let $X = 1 + s/D$. Then: $F_{1-2} = F_{2-1} = \frac{1}{\pi} \left[\sqrt{X^2 - 1} + \sin^{-1} \frac{1}{X} - X \right]$
7. 	$F_{1-2} = 1$, $F_{2-1} = \frac{r_1}{r_2}$, and $F_{2-2} = 1 - F_{2-1} = 1 - \frac{r_1}{r_2}$

Adapted From Lienhard & Lienhard "A Heat Transfer Textbook" (2012)

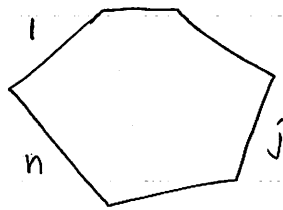
Table 10.3 View factors for some three-dimensional configurations

Configuration	Equation
1. 	Let $X = a/c$ and $Y = b/c$. Then: $F_{1-2} = \frac{2}{\pi XY} \left\{ \ln \left[\frac{(1 + X^2)(1 + Y^2)}{1 + X^2 + Y^2} \right]^{1/2} - X \tan^{-1} X - Y \tan^{-1} Y + X\sqrt{1 + Y^2} \tan^{-1} \frac{X}{\sqrt{1 + Y^2}} + Y\sqrt{1 + X^2} \tan^{-1} \frac{Y}{\sqrt{1 + X^2}} \right\}$
2. 	Let $H = h/\ell$ and $W = w/\ell$. Then: $F_{1-2} = \frac{1}{\pi W} \left\{ W \tan^{-1} \frac{1}{W} - \sqrt{H^2 + W^2} \tan^{-1} (H^2 + W^2)^{-1/2} + H \tan^{-1} \frac{1}{H} + \frac{1}{4} \ln \left[\frac{(1 + W^2)(1 + H^2)}{1 + W^2 + H^2} \right] \times \left[\frac{W^2(1 + W^2 + H^2)}{(1 + W^2)(W^2 + H^2)} \right]^{W^2} \left[\frac{H^2(1 + H^2 + W^2)}{(1 + H^2)(H^2 + W^2)} \right]^{H^2} \right\}$
3. 	Let $R_1 = r_1/h$, $R_2 = r_2/h$, and $X = 1 + (1 + R_2^2)/R_1^2$. Then: $F_{1-2} = \frac{1}{2} \left[X - \sqrt{X^2 - 4(R_2/R_1)^2} \right]$
4. 	Concentric spheres: $F_{1-2} = 1, \quad F_{2-1} = (r_1/r_2)^2, \quad F_{2-2} = 1 - (r_1/r_2)^2$

Now we can formulate our analysis as:

$$q_{12} = \frac{A_1 F_{12} (e_{b1} - e_{b2})}{A_1 F_{12} e_{b1} - A_2 F_{21} e_{b2}} \Leftrightarrow \begin{array}{c} \xrightarrow{\frac{1}{A_1 F_{12}}} \\ e_{b1} \xrightarrow{q_{12}} e_{b2} \end{array}$$

If we have an enclosure: (Radiation exchange inside a body)

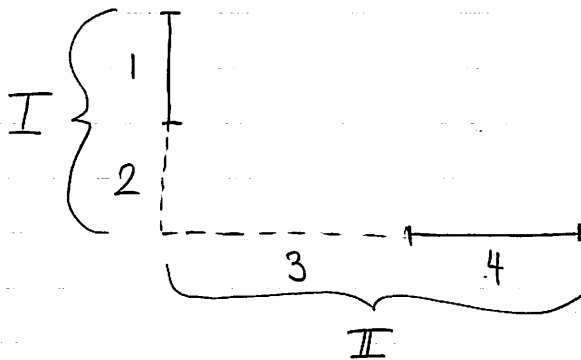


$$\sum_{j=2}^n F_{ij} = 1$$

Note, only true for flat or convex surfaces. If concave

$\sum_{j=2}^n F_{ij} \neq 1$ since $F_{ii} \neq 0$ (Surface sees itself).

Example Consider the following: (Solve for F_{14})



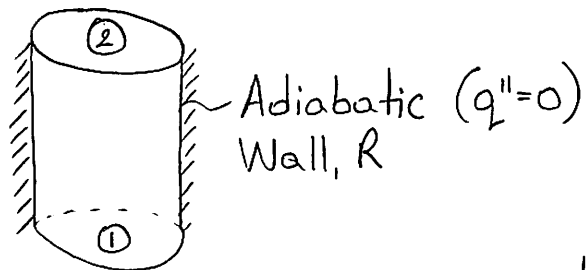
$$A_1 F_{14} = A_I F_{I4} - A_2 F_{24}$$

$$\cancel{A_I} F_{I4} = \cancel{A_I} F_{I\text{II}} - \cancel{A_I} F_{I3} \Rightarrow F_{I4} = \underbrace{F_{I\text{II}} - F_{I3}}$$

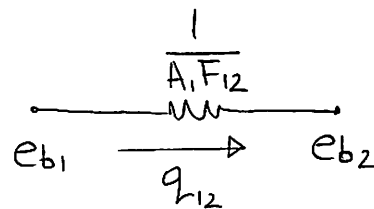
$$\cancel{A_2} F_{24} = \cancel{A_2} F_{2\text{II}} - \cancel{A_2} F_{23} \Rightarrow F_{24} = \underbrace{F_{2\text{II}} - F_{23}}$$

Our charts have these factors: $F_{I\text{II}}, F_{I3}, F_{2\text{II}}, F_{23}$ Can easily look these up. (211)

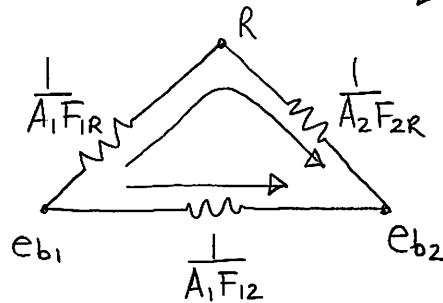
Adiabatic Surfaces (no heat transfer through them)



Previously we had:



Now we have (due to $q''=0$)



To solve, we now have:

$$\begin{aligned}
 & \left(\frac{1}{A_1 F_{1R}} \quad \frac{1}{A_2 F_{2R}} \right) \} R_{II} \\
 & \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \end{array} \\
 & \text{eb}_1 \quad \underbrace{\frac{1}{A_1 F_{12}}}_{R_I} \quad \text{eb}_2
 \end{aligned}
 \Rightarrow \frac{1}{R_{TOT}} = \frac{1}{R_I} + \frac{1}{R_{II}}$$

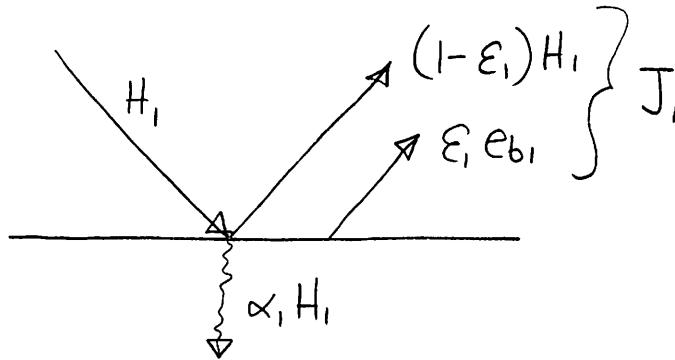
$$= A_1 F_{12} + \frac{1}{\frac{1}{A_1 F_{1R}} + \frac{1}{A_2 F_{2R}}}$$

$$q_{12} = \frac{e_{b1} - e_{b2}}{R_{TOT}} = (e_{b1} - e_{b2}) \left[A_1 F_{12} + \frac{1}{\frac{1}{A_1 F_{1R}} + \frac{1}{A_2 F_{2R}}} \right]$$

Gray Body Radiation

If $\epsilon \neq 1$, and $\epsilon \neq f(\lambda, \theta)$, we call this a gray body. Here, we can assume that $\epsilon = \alpha$.

← diffuse & independent of wavelength



Flux of energy that irradiates the surface.
 $H_i = \text{irradiance} \left[\frac{W}{m^2} \right]$

$J_1 = \text{radiosity} \left[\frac{W}{m^2} \right]$

↳ Total flux of radiat energy away from the surface.

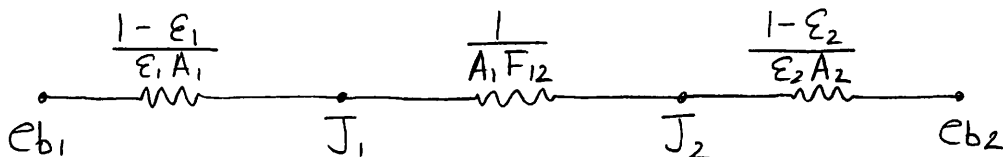
$$J_1 = \underbrace{(1 - \epsilon_1)H_i}_{\text{Reflected energy}} + \underbrace{\epsilon_1 e_{b1}}_{\text{Emitted energy}}$$

$$H_i = \frac{J_1 - \epsilon_1 e_{b1}}{1 - \epsilon_1}$$

$$\begin{aligned} \frac{q_{1, \text{NET}}}{A_1} &= (J_1 - H_i) = J_1 - \frac{J_1 - \epsilon_1 e_{b1}}{1 - \epsilon_1} \\ &= \frac{J_1 (1 - \epsilon_1) - J_1 + \epsilon_1 e_{b1}}{1 - \epsilon_1} = \frac{\epsilon_1}{1 - \epsilon_1} (e_{b1} - J_1) \end{aligned}$$

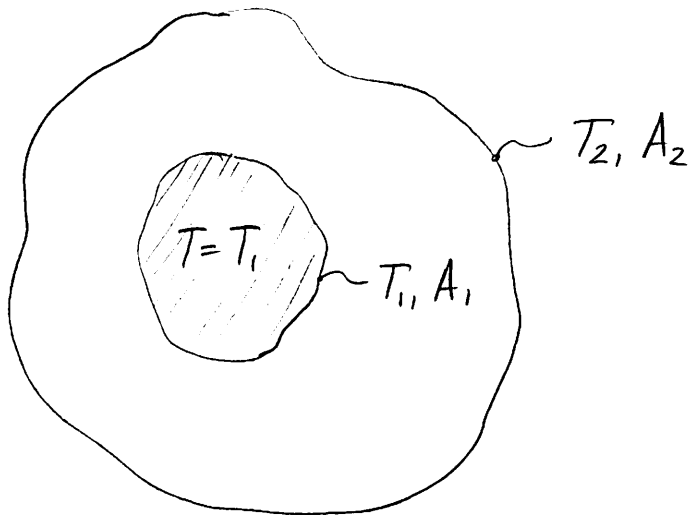
$$\boxed{\frac{q_{1, \text{NET}}}{A_1} = \frac{\epsilon_1}{1 - \epsilon_1} (e_{b1} - J_1)} \Rightarrow \text{Net heat flux into surface 1.}$$

We can reformulate this in terms of a thermal resistance



Note, if you set $\epsilon_1 = \epsilon_2 = 1$, you get the blackbody solution.

Gray Body Enclosures



By an energy balance:

$$A_1 H_1 = A_2 J_2 F_{2,1} + A_1 J_1 F_{1,1}$$

$$\quad \quad \quad \underbrace{A_1 J_2 F_{2,1}}_{A_1 J_2 F_{12}}$$

$$H_1 = J_2 F_{12} + J_1 F_{1,1}$$

↳ Net energy coming to ①

$$\frac{q_{1 \rightarrow 2, \text{NET}}}{A_1} = J_1 - J_2 F_{12} - J_1 F_{1,1}$$

$$= \underbrace{(J_1 - J_2) F_{12}}_{\text{Since } (1 - F_{1,1}) = F_{12}}$$

So we have the same solution as before, but assuming that body 2 is much larger than body 1, we get $F_{12} = 1$.

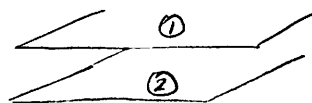
$$q_{1 \rightarrow 2} = \frac{A_1 (e_{b1} - e_{b2})}{\left(\frac{1}{\epsilon_1} - 1\right) + \frac{1}{F_{12}} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1\right)}$$

① $A_2 \gg A_1$; $F_{12} = 1$

$$q_{1 \rightarrow 2} = A_1 \epsilon_1 (e_{b1} - e_{b2}) \Rightarrow \text{Smaller body behaves like a black body.}$$

Physically, this result comes from the fact that the emitted and reflected radiation from ① bounces around multiple times and is absorbed by ② before it can hit ① again. Hence $\epsilon_2 = 1$.
 $\alpha_2 = 1$.

② Parallel Plates:



$$F_{12} = 1 = F_{21}$$

$$A_1 = A_2$$

$$q_{1 \rightarrow 2} = \frac{A (e_{b1} - e_{b2})}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1}$$